

IS105 - ENTERPRISE ARCHITECTURE

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INTRODUCTION TO ENTERPRISE ARCHITECTURE



AT THE END OF THIS LESSON, YOU SHOULD BE ABLE TO:

- Discuss the fundamental concepts and applications of Enterprise Architecture.
- Describe the interrelationships among Business Architecture, Information Systems (Data and Applications) Architectures, and Technology Architecture.
- Be conscious how how an effective Enterprise Architecture can enable an organization to accomplish its business goals.

ENTERPRISE

ARCHITECTURE

- The ISO/IEC 42010 International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC). standard defines architecture.

ENTERPRISE

ARCHITECTURE

- It is the fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles governing its design and evolution.
- In this context the system that is under discussion is the enterprise

ENTERPRISE

ARCHITECTURE

- It recognizes the complexity of modern enterprises, which encompass human, organizational, and technological elements.
- Its primary goal is to manage and align an organization's IT infrastructure with its business objectives.

ENTERPRISE ARCHITECT (PROGRAM APPLICATION)



ENTERPRISE

ARCHITECT

- **Enterprise Architect** is used to define or document the description of the architecture and therefore provides a representation or way of visualizing a planned or built architecture.
- It is a popular software application used to visualize and manage complex organizational structures.

ENTERPRISE

ARCHITECT

- Modeling - Creating visual representations of business processes, data flows, and IT systems.
- Alignment - Demonstrating how IT initiatives support strategic goals.
- Impact analysis - Assessing the potential consequences of changes to systems or processes.

LEVELS OF ARCHITECTURE



LEVEL OF

- **Big companies are like big puzzles.** To put them together correctly, you need different sized pieces. These pieces are called "architectures".
- Just like big puzzles have big, medium, and small pieces, companies need big, medium, and small plans for their computers and systems.
- These plans should match the company's

ARCHITECTURE

LEVELS OF

- An Enterprise has a complex and typically hierarchical structure and architectures will need to be created at discrete levels of this structure.
- This hierarchy of architectures is analogous to the hierarchies of goals and capabilities and intuitively aligns with Strategic, Program and Project level divisions.

LEVELS OF

ARCHITECTURE
The naming of the levels has been influenced by The Open Group Architecture Framework (TOGAF).

- **Strategic** - Long term in the range of 3 - 5 years.
- **Tactical** - Mid term in the range of 1 - 2 years
- **Solution** - Short term in the range of 6 - 12

Months

TYPES OF ARCHITECTURE



TYPES OF ARCHITECTURE

The overall architecture of an enterprise can be described by four integrated sub-architectures.

These are:

- Business Architecture
- Information Architecture
- Application Architecture
- Technology Architecture

TYPES OF

ARCHITECTURE

There are a number of other architectures that could best be described as views, as they cross-cut and are contained by the other architectures, but because of their importance they are often raised to the level of an architecture.

These include:

- Security Architecture
- Geospatial Architecture
- Social Architecture

BUSINESS

ARCHITECTURE

- The business architecture will typically consist of a description of the baseline and target architectures, with a series of transitions defined that can be executed and that would be described on Roadmap diagrams.

BUSINESS

ARCHITECTURE

- It defines the business drivers, the business strategy, operational models, goals and objectives that the organization needs to achieve to transition in a potentially competitive and disruptive business environment.

« mission »

Our Customers are our priority and we will provide them with first class service and products in a timely manner and listen to their needs attentively. We achieve our excellence through the tireless efforts of our staff that we support enthusiastically.



« vision »

To be the leader in the delivery of knowledge products to customers wherever they are in the world.

This Diagram shows how a Mission and Vision Statement can be modeled visually. The element visualization has been changed to use Rectangle Notation allowing the Stereotypes to be displayed in the diagram.

INFORMATION

ARCHITECTURE

- Information Architecture is key to the success of an Enterprise Architecture Program as information is created consumed and destroyed by the components that make up the other architectures.

INFORMATION

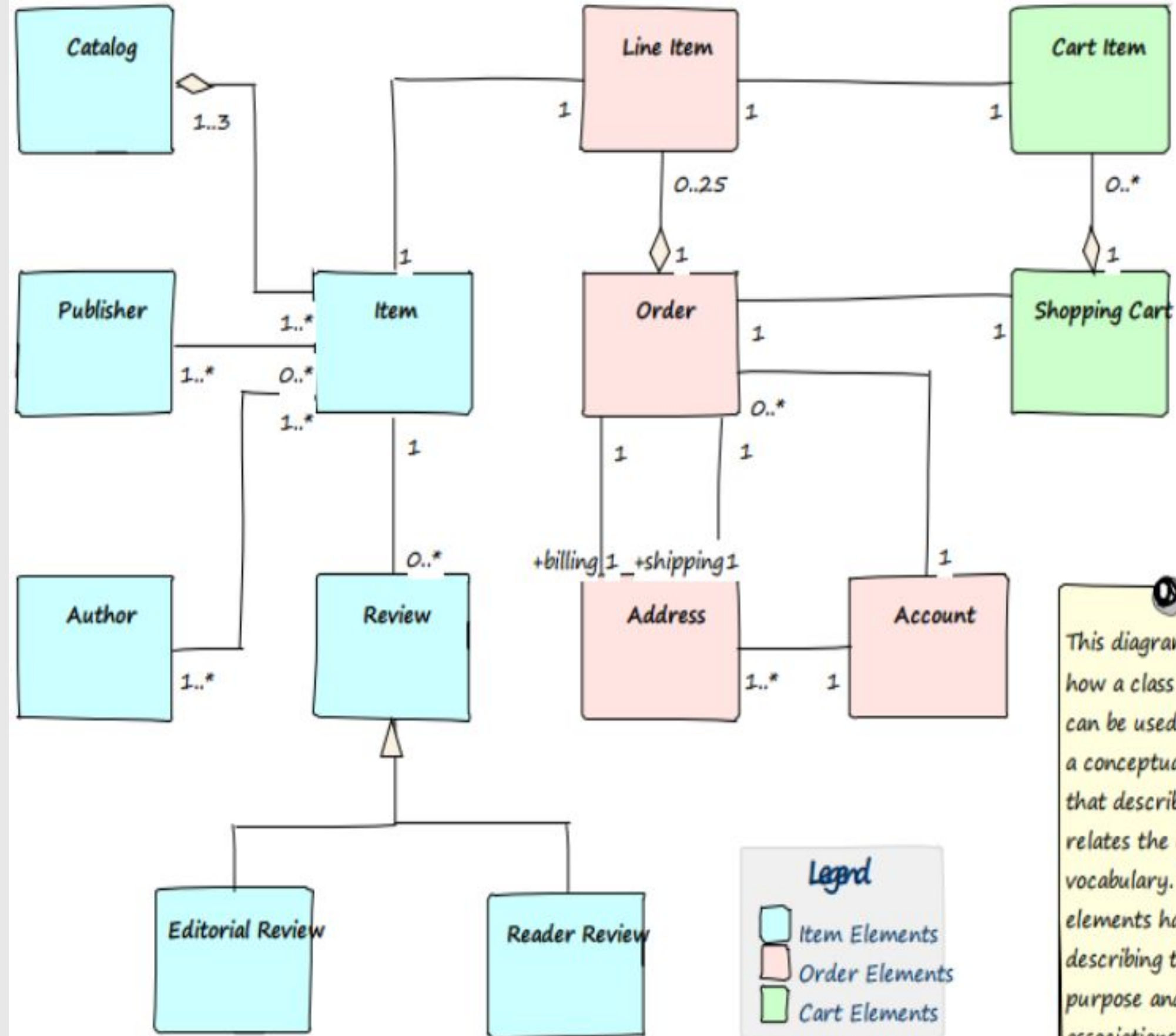
ARCHITECTURE

- Understanding which business functions and processes utilize information, which applications serve as the master record, where information is created and destroyed and which technology components store and manipulate the information is critical to achieving the business outcomes.

INFORMATION

ARCHITECTURE

- The information architecture will typically consist of a description of the baseline and target architectures with a series of transitions defined that can be executed and which would be described on roadmap diagrams.



This diagram shows how a class diagram can be used to create a conceptual model that describes and relates the business vocabulary. The elements have notes describing their purpose and associations relationships and multiplicities describe how they relate to other business terms.

Application Architecture

- The application architecture provides an important catalog of the applications in the enterprise describing the work that they do to transform, transmit and store information.

Application Architecture

- The architecture also describes the interfaces that are required or provided by the applications and the way the applications interact to carry out the activities described in the business models such as the business process diagrams and capability models.

Application Architecture

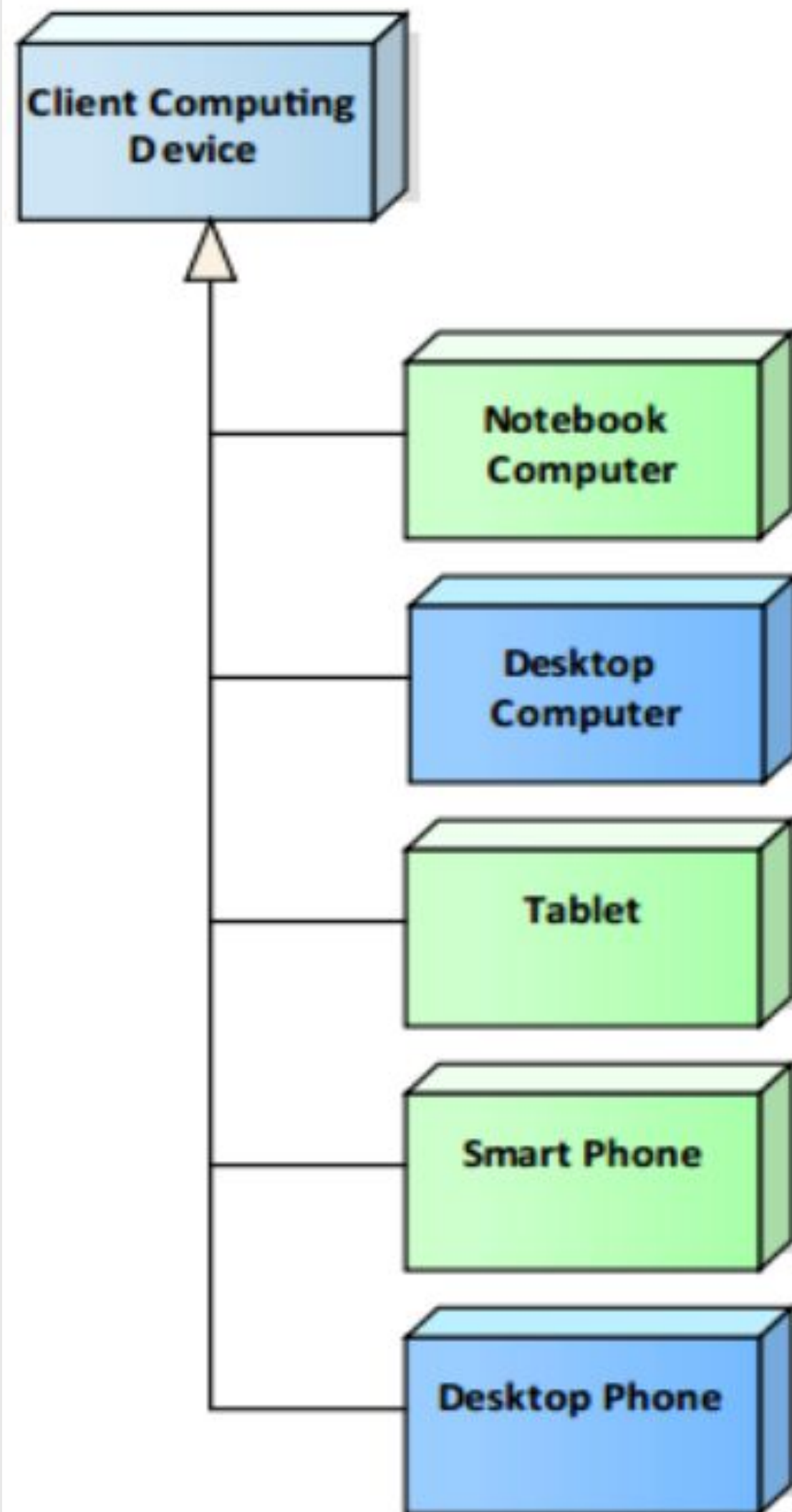
- The catalog of applications, interfaces and the diagrams and matrices that describe their interaction only need to be defined once at the enterprise level.



This Component diagrams shows the details of the interface between two Logical Components using Ports and Interfaces.

Technology Architecture

- The technology architecture underpins the other architectures providing a description of the logical, physical and virtual infrastructure that supports the execution of application services which in turn support information and business functions and services.



This Deployment diagram shows how a hierarchy of Client Computing Devices elements arranged vertically using a lateral vertical line style.

Portability

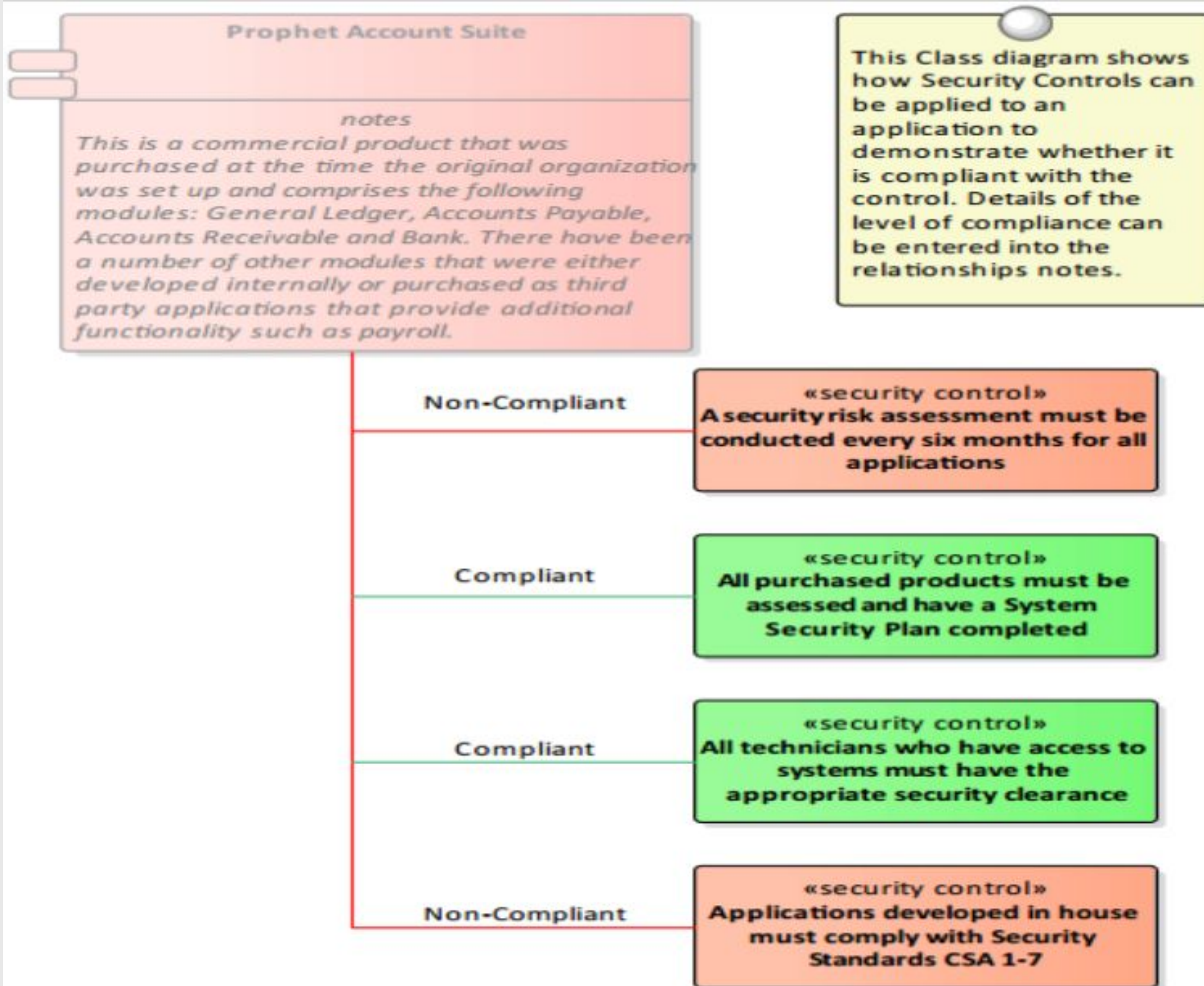
-  Portable
-  Fixed

Security Architecture

- The security architecture is a slice through all of the other architectures from a security viewpoint.
- It is listed as a separate architecture because of its importance in ensuring that the enterprise security policies are implemented through the architecture.

Security Architecture

- A breach of security could occur at any point from the business architecture through to the technology architecture.
- This could include demonstrating how the architectures comply with security controls published by the enterprise or available as part of an industry compliance regulation.



Geospatial Architecture

- The Geospatial Architecture is a slice through all of the other architectures from a geo-spatial or location based viewpoint.
- It is listed as a separate architecture because of its increasing importance in a world dominated by location based applications and business technology functions.

Social Architecture

- Social Architecture has been born in an era that depends on social media and social behavior to achieve a wide range of outcomes in built and digital environments.
- Increasingly in this period of digital and social disruption, understanding the social aspects of an organization's interaction is critical.

Social Architecture

- This can include the way it interacts with its customers, suppliers and observers and other individuals and communities through what is commonly described as social media.

SCOPE OF ARCHITECTURE



Scope of Architecture

- Architectures will only be successful if they are scoped correctly. The Enterprise Architecture Body of Knowledge (EABOK) describes three important aspects of scope but another one can be added that addresses the importance of the stakeholders in the success of the architecture program and the architectures it creates and manages:

Scope of Architecture

- Time Scope
- Organization Scope
- Detail Scope
- Stakeholder scope

Architecture time frames, organizational context, levels of detail and appeal to stakeholders must all be appropriately set for the architecture to be relevant and successful.

Time Scope

- Time scope is crucial in Enterprise Architecture (EA). It means that different plans need different amounts of time to be completed.
- This helps companies stay organized and make sure they're working towards their long-term goals.

Time Scope

- It's like planning a vacation. You start with a big plan of where to go (strategic), then plan specific trips (tactical), and finally, plan daily activities (solution).
- All these plans need to fit together smoothly.

Detail scope

- Selecting the correct level of detail for an architecture is critical to its success and this is particularly true when it comes to the Implementation teams.
- Creating architectures that are too lofty or that are aspirational will result in Implementation teams making important design decisions themselves.

Detail scope

- On the flip side of this argument creating architectures that are too prescriptive and detailed will often result in an Implementation team being constrained and often results in the team not having the flexibility to select the best solution.

Organization Scope

- Having a clear understanding of the structure of an enterprise and its organizations and how strategic plans relate to this structure is critical for the success of any enterprise architecture effort.

Executive
Management

This Organization Chart diagram is part of the Strategic Modeling extension and allows organizational structure to be modeled, either at an organizational unit level or at a role level.

Finance

Portfolio
Management

Sales

Research and
Development

Production and
Logistics

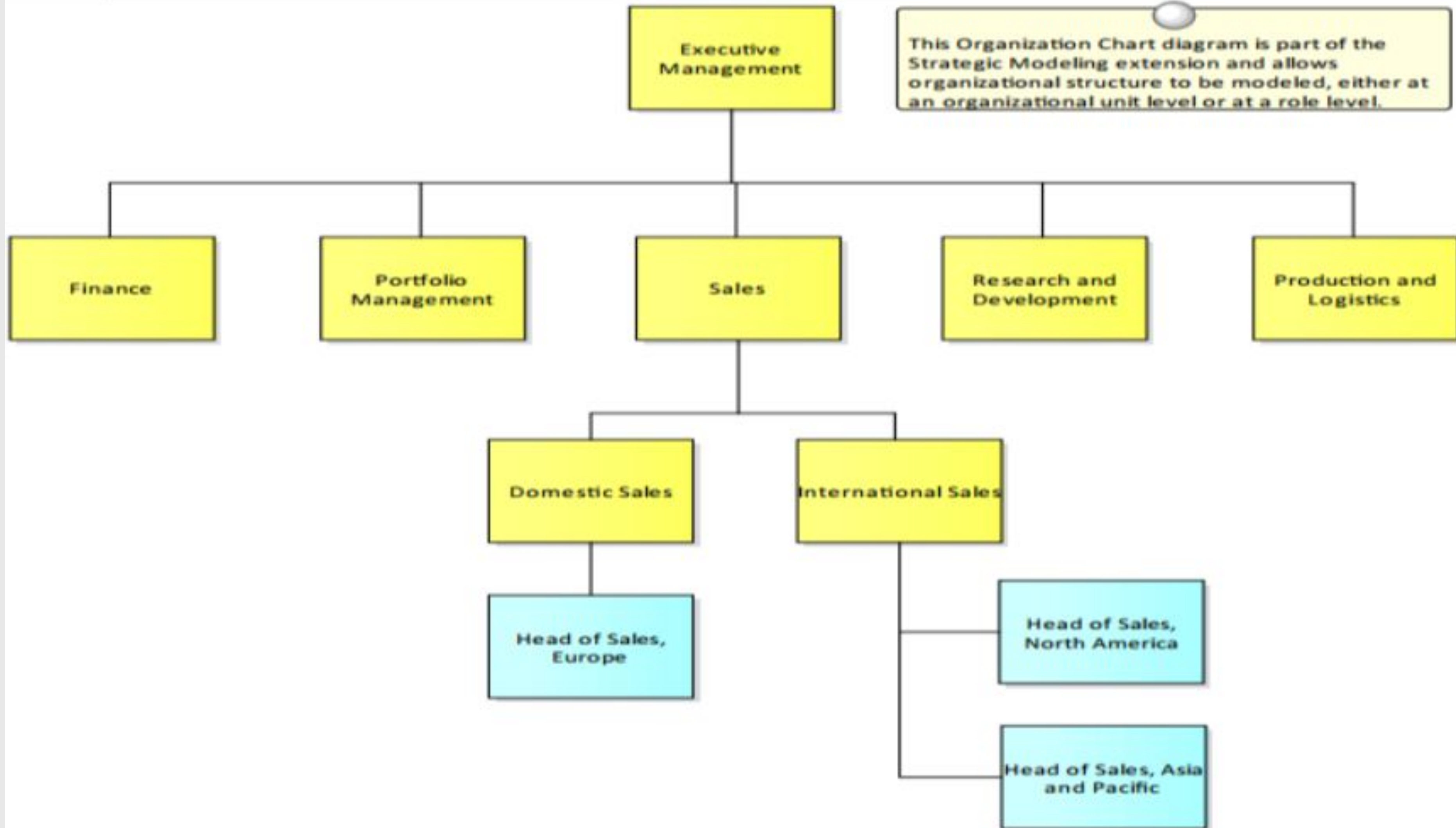
Domestic Sales

International Sales

Head of Sales,
Europe

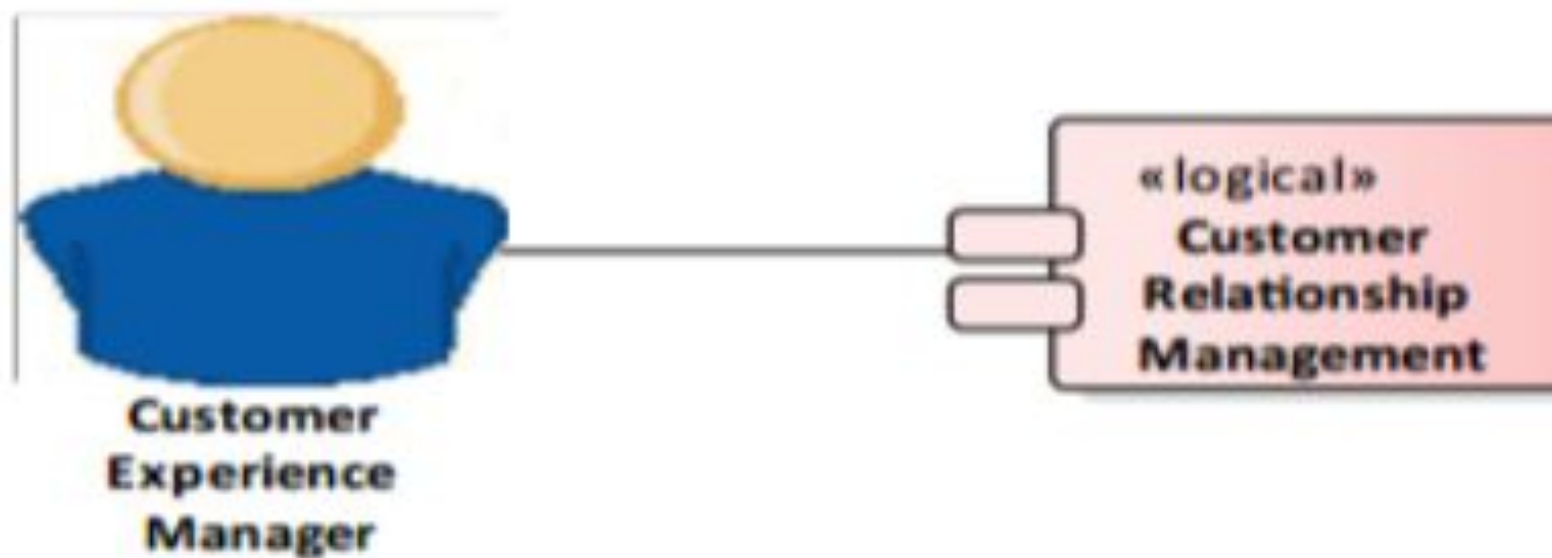
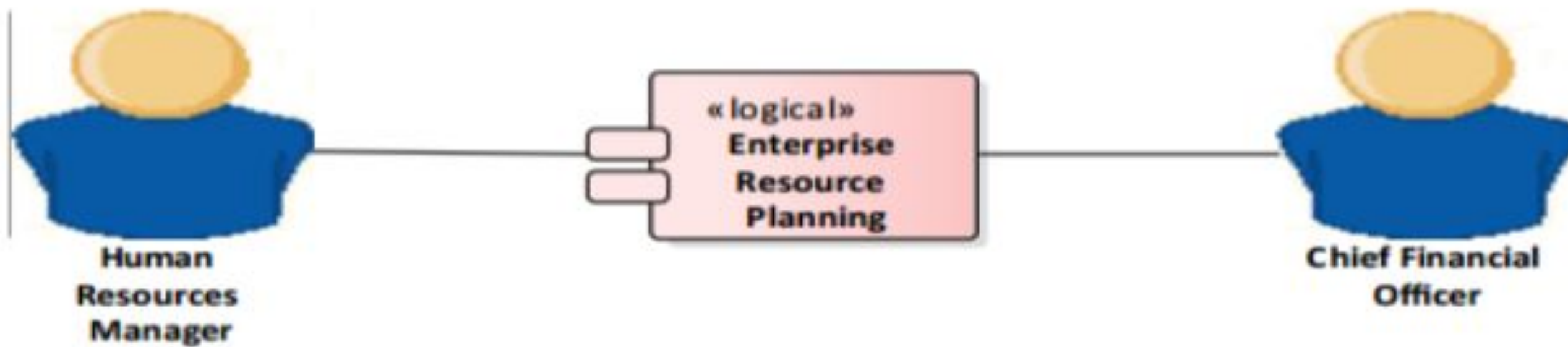
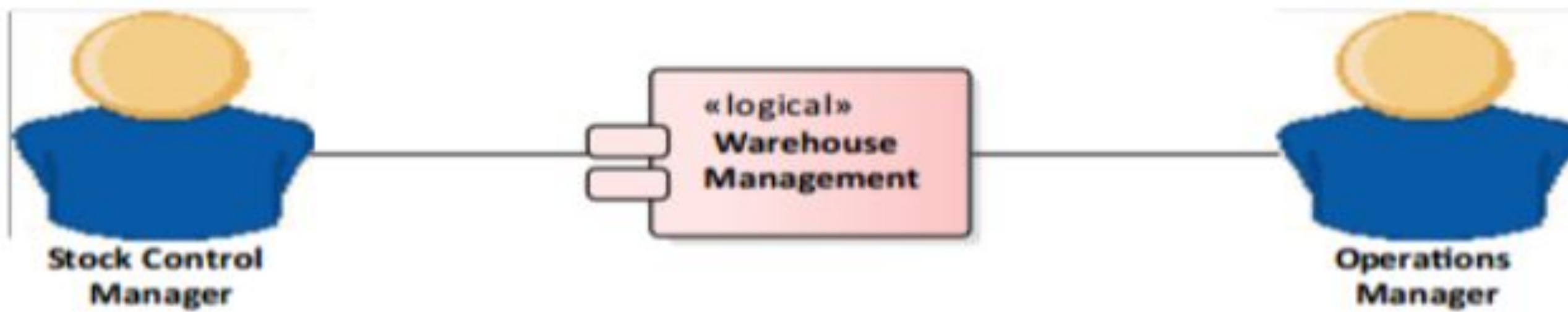
Head of Sales,
North America

Head of Sales, Asia
and Pacific



Stakeholder Scope

- The stakeholders and the shareholders or organization owners that they represent are the ultimate beneficiaries of the enterprise architecture and it is important that the right stakeholders are selected and communication is managed to ensure they are kept updated with the progress of the architectural work and the governance of the Implementation Initiatives.



This diagram shows the relationship between stakeholders and solution components using an Association relationship. The stakeholders are UML Classes that have had an image assigned as an appearance option.

CHARACTERISTICS OF GOOD ARCHITECTURE



Characteristics of Good Architecture

- It is difficult to define what the characteristics of good architecture are when there is still vigorous and lasting debate about what the term architecture actually means in the context of enterprise systems in the twenty first century.

Characteristics of Good Architecture

- The Roman architect Vitruvius defined three characteristics of good architecture in his treatise *De Architectura* more than 2,000 years ago. Interestingly it is the only surviving text from antiquity describing architecture.

Characteristics of Good Architecture

- Durability (Firmatis) – It should stand up robustly and remain in good condition
- Utility (Utilitas) – It should be useful and function well for the people using it
- Beauty (Venustatis) – It should delight people and raise their spirits

QUALITIES OF GOOD ARCHITECTURE



Qualities of Good Architecture

- To be effective, an architecture must have a number of qualities or characteristics.
- It makes sure technology works well for the business, is flexible, easy to understand, cost-effective, and secure.

Qualities of Good Architecture

- **Utilitarian** - The architecture serves a practical purpose and fulfills its intended function.
- **Durable** - The architecture is built to last and withstand the test of time.
- **Flexible** - It can adapt to changing needs and circumstances.

Qualities of Good Architecture

- **Verifiable** - The architecture can be evaluated and validated to ensure it meets requirements.
- **Elegant** - The architecture is aesthetically pleasing and efficient.
- **Traceable** - The architecture can be traced back to its origins and decisions.

THE END